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# The impact of market and policy instability on price transmission between wheat and flour in Ukraine

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## Abstract

Analysis of price transmission in transition countries is often complicated by unstable policy environments. We use a Markov-switching vector error-correction model (MSVECM) to model multiple regime shifts in the relationship between wheat and wheat flour prices in Ukraine from June 2000 to November 2004. Unlike common alternative error correction specifications, the MSVECM does not require an explicit transition variable but rather permits changes between regimes to depend on an unobserved state variable. The analysis reveals four regimes whose timing coincides with political and economic events in Ukraine. Strong coincidence between a 'high-uncertainty' regime and discretionary policy interventions suggests that policy responses to fluctuations in Ukrainian harvests may have amplified instability.

**Keywords:** Markov-switching vector error-correction model, vertical price transmission, regime shifts, grain policy, Ukraine

**JEL classification:** Q11, Q18, P22, P23

## 1. Introduction

> Oh! I will go into business again, I will buy wheat in Odessa; out there, wheat fetches a quarter of the price it sells for here. There is a law against the importation of grain, but the good folk who made the law forgot to prohibit the introduction of wheat products and food stuffs made from corn. Hey! Hey!? That struck me this morning. There is a fine trade to be done in starch.
>
> – Honoré de Balzac, Old Goriot

Review coordinated by Alison Burrell

Prices play a key role in market economies. They coordinate the decisions of producers and consumers in a manner that, under conditions of perfect competition, leads to a *Pareto* optimal allocation of scarce resources. While the conditions that define perfect competition are never met to the letter, the efficiency of the price mechanism as a means of allocation under a broad range of realistic conditions is widely acknowledged.

The price mechanism was not allowed to play this role in the centrally planned economies of Central and Eastern Europe (CEE). At the onset of transition, hopes were therefore pinned on market liberalisation and the harnessing of the price mechanism as a means of increasing economic efficiency and welfare. Consequently, economists have invested much effort in monitoring market liberalisation and the functioning of emerging markets in CEE countries. An important strand of the resulting literature deals with price transmission and the integration of agro-food markets in the CEE region (Goodwin *et al.*, 1999; Loy and Wehrheim, 1999; Berkowitz and DeJong, 2000; Kuhn, 2000; Bojnec and Peter, 2002; Bakucs and Fertő, 2005). We add to these studies by analysing the vertical transmission from milling wheat to flour prices in Ukraine.

The literature on price transmission and market integration in CEE is part of a much larger and rapidly growing body of literature on price transmission that highlights both the development of new empirical methods and their application in various product and country settings. Economists have developed a variety of empirical methods for studying price transmission and market integration (see, for example, Fackler and Goodwin, 2001). In the course of this development, the study of simple correlations between price series has been supplanted by increasingly sophisticated econometric techniques. The introduction of co-integration methods in the 1980s gave impetus to the price transmission literature by enabling practitioners to distinguish non-spurious from spurious relationships between (commonly non-stationary) prices, and by providing deeper insights into the equilibrating dynamics – generally attributed to arbitrage – that underlie the former.

In the late 1990s, however, research revealed pitfalls associated with using co-integration methods to analyse price transmission and market integration (Baulch, 1997; McNew and Fackler, 1997; Barrett, 2001). For example, failure to account for non-stationary transfer costs (e.g. transport costs in a spatial context, or marketing margins in a vertical context) when analysing price co-integration can generate misleading results. Similarly, reversals of trade flows due to supply or demand shocks can lead to price series that are not co-integrated, even though the markets in question are integrated.

These insights have spurred economists to refine further the empirical methods used to analyse price transmission. Several lines of work can be identified, all of which are based on non-linear models that allow for regime-specific behaviour in the relationship between the prices being studied. First, Goodwin and Piggott (2001), Meyer (2004) and Sephton (2003), among others, have adapted threshold error correction techniques to the study of price transmission.[1] Second, in the asymmetric price transmission literature, regime shifts depend on whether prices are increasing or decreasing. Meyer and von Cramon-Taubadel (2004) reviewed this literature, and Chavas and Mehta (2004) developed a general model that nests many earlier approaches. Third, Quandt's (1958) switching regression model has been elaborated to allow for regime-specific behaviour in the long-run relationship between prices. A prominent example is smooth transition regression modelling (e.g. Teräsvirta, 2004, and the literature cited therein). Along related lines, Hansen (1992) and Gregory and Hansen (1996) derived tests for structural breaks in co-integrating relationships, while Gonzalo and Pitarakis (2006) offered a test for threshold effects in co-integrating relationships. Finally, elaborating on the parity bounds model (PBM) developed by Baulch (1997), Barrett and Li (2002) employed a mixture distribution model that incorporates not only prices but also transfer cost and trade flow data.

These modelling approaches have broadened the set of empirical tools available and deepened our understanding of price transmission. Nevertheless, several issues remain to be addressed. An important issue for the empirical application to wheat and flour prices in Ukraine explored here is that, in all these modelling approaches, regime shifts are driven by observable transition variables that are explicitly included in the empirical analysis. In threshold error-correction models, regime shifts are determined by the size of the error correction term (i.e. the deviation from long-run equilibrium) relative to some threshold value. In the analysis of asymmetric price transmission, the transition variable is an indicator function that signals whether prices are rising or falling. In smooth transition regression, various transition variables are possible, such as error correction terms and changes in GNP (e.g. Teräsvirta, 2004), and time trends (e.g. Goodwin *et al.*, 2000). In the PBM, six possible regimes are defined by two possible states of trade between two locations (takes place or not) multiplied by three possible states of the difference in price between these locations (greater than, equal to or less than transfer costs between the locations).[2]

In the case of wheat–flour price transmission in Ukraine over the 2000–2004 period, we expect regime-specific behaviour because Ukraine twice shifted from a net export to a net import position and back for milling wheat. Furthermore, policy makers in Ukraine, partly in reaction to these shifts, frequently intervened on domestic wheat and flour markets, sometimes severely affecting the policy environment within which traders and millers operate and interact. The resulting changes in the behaviour of traders and

1 These models are sometimes referred to as 'threshold co-integration' models although, as Gonzalo and Pitarakis (2006) pointed out, this name is misleading here. These models assume that there is a unique long-run relationship between the prices studied, but allow the equilibrating dynamics that maintain this relationship to change depending on the size of the error-correction term with respect to some threshold value(s). Hence, it is not the co-integration itself that is subject to threshold effects but rather the adjustment mechanism.

2 Serra *et al.* (2006) used non-parametric techniques to analyse price transmission. Although their approach also requires an observable transition variable, its main advantage is that it allows the response to this transition variable to be more flexible than in parametric models.

millers, and thus in price transmission between wheat and flour, cannot be linked to a specific, observable transition variable. Regime changes in our application are not driven by an error correction term or the direction of price changes but rather by variables such as expectations and policy interventions that are not directly observable, or at least very difficult to measure reliably. The PBM is also inappropriate in our setting because there are no data available on weekly flows of wheat to millers that would correspond to the weekly price data at our disposal. Indeed, while the 'hazards of omitting trade volumes and transactions costs' (Barrett, 2001: 24) in price transmission analysis are manifest, in many interesting settings these data are simply not available, precluding the use of the PBM.[3] An additional limitation of the PBM is that it fails to take the time series nature of price, trade and transaction cost data into account. Instead, it attributes each observation in a series individually to one of the six possible regimes using maximum likelihood methods. Failure to account for the time series nature of data (e.g. when prices in one period affect trade flows in subsequent periods) can reduce the efficiency of estimation, and greatly limits the insights into the dynamics of price transmission that the PBM can provide.

In summary, none of the non-linear modelling approaches discussed above is appropriate for the analysis of wheat–flour price transmission in Ukraine. For this reason, we chose an approach that is new to the price transmission literature – the Markov-switching vector error-correction model (MSVECM). The MSVECM allows for regime-specific price transmission behaviour, where regime shifts are determined by an unobserved state variable. It provides insights into the temporal dynamics of price response in the vertical wheat–flour chain, and can be implemented with price data alone.

The rest of the paper is structured as follows. Section 2 provides an overview of policy developments on Ukrainian wheat and flour markets, explaining why we expect regime-specific price transmission between wheat and flour in Ukraine, and what triggers regime shifts. Section 3 introduces the MSVECM and outlines why it is suited to deal with the sort of regime-specific price transmission that we expect on Ukrainian wheat and flour markets. A description of our data is also given in Section 3. Empirical results are presented in Section 4, and are discussed in Section 5 in relation to developments on Ukrainian wheat and flour markets. Section 6 concludes.

## 2. Policy intervention on wheat and wheat flour markets in Ukraine

Policy makers in Ukraine actively intervene in agricultural markets. This intervention combines elements of Soviet-style agricultural policy (such as

3 That said, priority should be given to generating such complete data sets, as they have unique potential to cast new light on questions of market integration. The difficulty is collecting trade flow data at the same high (monthly, even weekly) frequencies at which price data are often available, and which are required to expose the intricate dynamics of interaction between markets.

state procurement) with features of the market and price support that characterises agricultural policy in many industrialised countries. The main forces driving this often highly contradictory policy mix have been: (i) very significant reductions in agriculture's terms of trade in Ukraine following the onset of transition, as in most transition countries (see, for example, Rozelle and Swinnen, 2004); (ii) high hidden unemployment and inefficiency of agricultural production, processing and marketing bequeathed by the Soviet era; (iii) a lack of the analytical capacity needed to appraise policies and their impacts in an open-economy setting; and (iv) rent-seeking by policy makers, entrepreneurs and other stakeholders in an environment characterised by information asymmetry, lack of transparency and weak legal and regulatory institutions (on the last three points, see von Cramon-Taubadel *et al.*, 2008).

Ukrainian grain markets, and wheat markets in particular, are highly politicised. Grain is considered 'strategic', and agricultural policy makers consider 'the size of the grain harvest [to be] a barometer of conditions in agriculture' (von Cramon-Taubadel, 2001: 103). Until the end of the 1990s, grain policy remained dominated by Soviet-style intervention in the form of so-called state orders. Government bodies and parastatal enterprises supplied key inputs for autumn and spring sowing campaigns to the large collective farms, most of which, although renamed, had undergone little meaningful restructuring. In return, the authorities collected grain in payment for these inputs after the harvest (Striewe and von Cramon-Taubadel, 1999). In keeping with the adage 'you pretend to pay us, and we pretend to work', the inputs supplied were generally too few, of poor quality and often late, and in return the farms generally failed to deliver the contracted grain. Because large farms could not be permitted to go bankrupt, non-payment proliferated, leading to recurrent debt write-offs. Periodic bans on exports and movements of grain between oblasts (the main federal sub-unit, of which there are 26) within Ukraine were imposed in an effort to force farms to deliver more grain to the state and its agents. Although they affected farms unevenly and destabilised grain markets considerably, the state orders maintained soft budget constraints and permitted many farms that would otherwise have been forced to exit the sector to continue operations.

The state orders also crowded out private investment in the input supply and grain marketing industries and propped up a variety of state and parastatal monopolies in these industries. Cash-strapped farms could only purchase inputs against the next harvest. However, private input suppliers and grain traders, knowing that the state would attempt (via bans and controls on exports and movements of grain) to lay first claim on whatever grain was produced, were reluctant to provide credit on this basis. Thus, the state-order system stifled competition and investment in the industries up- and downstream from grain production. As a result, decayed and outdated grain marketing infrastructure was not modernised, and monopoly suppliers of grain storage, handling and transportation services were able to charge excessive prices, leading to very low farm-gate prices in the net export situation that prevailed up to 2000. According to one estimate, farms in Ukraine received only 40 per cent of the FOB export price for grain, compared with, for example, 70 per cent in Germany (Striewe and von Cramon-Taubadel, 1999). This indirect tax on grain production outweighed the direct budget subsidies referred to above, leading to a significant net taxation of grain production in Ukraine. Together with the other difficulties associated with transition, this contributed to a dramatic reduction in wheat production during the 1990s (Figure 1).

Following the financial crisis in 1998/1999, President Leonid Kuchma installed a new reform-oriented government under Prime Minister Viktor Yushchenko. In a key move, the new government eliminated state grain orders, reducing the state's direct influence on grain markets and fostering private enterprise in up- and downstream industries (von Cramon-Taubadel, 2001; Demyanenko and Zorya, 2004). However, because of bad weather in 2000, Ukraine's wheat harvest dropped to its lowest volume since independence, and the country became a net importer of wheat. The same excessive marketing costs that had depressed export parity prices in earlier years now inflated import parity prices, which more than doubled in the course of a few months (Figure 2).

As grain prices skyrocketed, bread prices became a major concern. The policy response was *ad hoc* and populist. The government accused traders of speculation to drive up wheat prices, introduced grain export certification, fixed bread prices and announced plans to introduce a 'pledge price' system based on the US loan rate system. To increase wheat supply, wheat import duties were temporarily suspended and the wheat import regime simplified.

High wheat prices, low supply and uncertainty about policy developments had an impact on flour producers. Domestic production of flour fell and flour

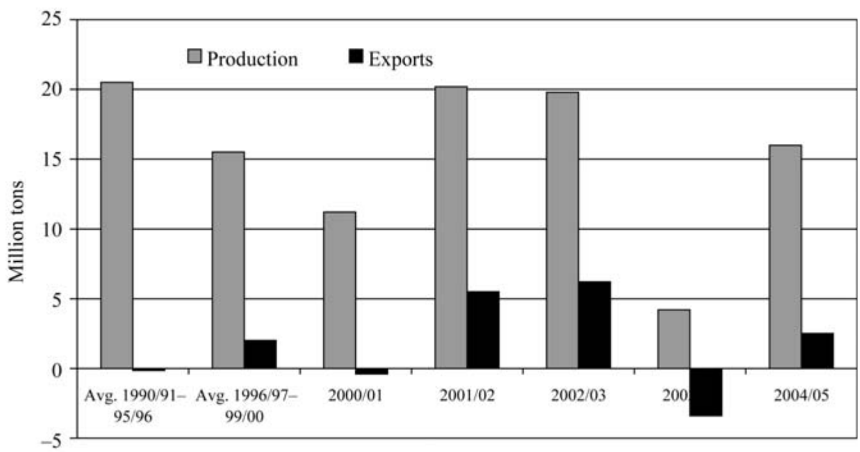


**Figure 1.** Production and net export of wheat in Ukraine, 1990–2005 (million tons).
*Source:* UkrAgroConsult (various issues) and USDA(a) (various issues).

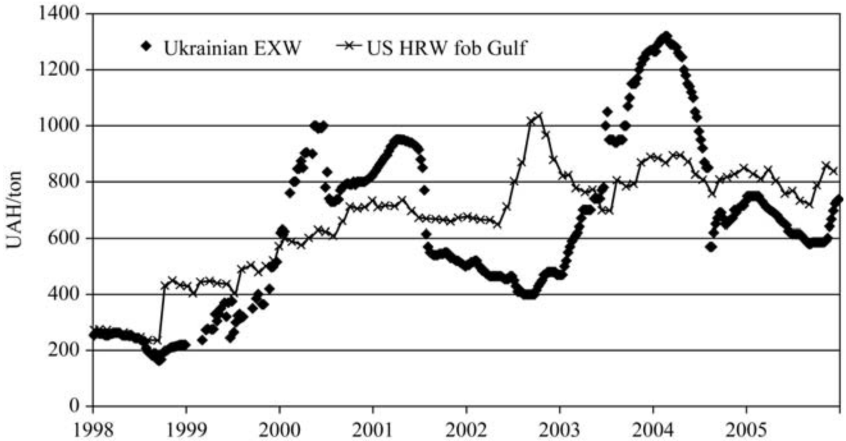


**Figure 2.** Ukrainian and world market wheat prices, 1998–2005 (UAH[4]/ton).
*Source:* UkrAgroConsult (various issues) and USDA(b) (various issues).
*Note:* Ukrainian EXW refers to milling wheat (class III) ex-warehouse price. US HRW fob Gulf refers to hard red winter free on board at the Gulf of Mexico and is used as an indicator of prices on the world market.

prices grew sharply, reaching new highs around 1,400 UAH/ton in the 2000/2001 marketing year. The evidence in Figure 3 suggests that flour producers attempted to take advantage of wheat price increases by increasing flour prices more than proportionally. Since bread prices were administratively fixed at low levels, bakers (officially) could not fully pass on flour price increases to consumers. However, anecdotal evidence suggests that a shadow market for bread flourished at the time, as supplies of bread at the fixed prices contracted significantly.

Responding to high wheat prices, reforms and favourable weather, the wheat harvest in 2001 reached its highest level in 10 years and remained high in 2002 (Figure 1). Ukraine became a major wheat exporter, exporting over 5 million tons of wheat in both 2001/2002 and 2002/2003. Domestic prices returned to low export parity levels, falling more than 40 per cent after the 2001 harvest as farms, needing to repay input credits and finance the autumn sowing campaign, were under pressure to sell quickly. Even where liquidity was less binding, storage was not an option since most farms had insufficient on-farm storage capacities and state-run elevators charged exorbitant fees. In response, the government reiterated its intention to implement the pledge price (loan rate) system. It also announced plans for an EU-style intervention system. Policy makers seemed unconcerned (or unaware) that one of these two systems was clearly redundant; in any event,

4 During the sample period for this analysis, the exchange rate between the Ukrainian hryvnia and the euro fluctuated considerably. It was EUR 1=UAH 5.1484 in June 2000, and EUR 1=UAH 7.1032 in December 2004.

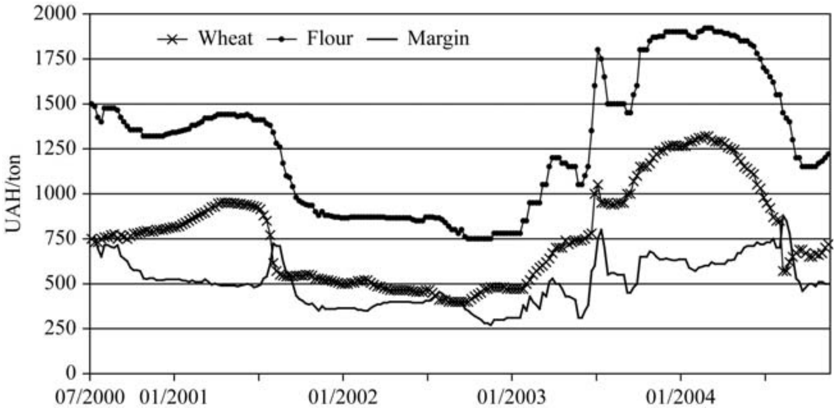


**Figure 3.** Weekly wheat and flour prices and the corresponding margin in Ukraine, July 2000–November 2004 (UAH/ton).
*Source:* UkrAgroConsult (various issues).
*Notes:* The wheat price is the same class III ex-warehouse price depicted in Figure 2. The flour price is the wholesale price for top-grade flour. The margin is the difference between these prices.

sufficient budget funds were never allocated for either. The government also attempted to regulate exports by requiring that export contracts be registered on official agrarian exchanges, but this requirement was not enforced evenly. On the positive side, steps were taken to reduce marketing costs and encourage private investment in market infrastructure, especially at sea ports and elevators. On the whole, however, policy remained 'stop-and-go' and a major source of uncertainty on Ukrainian wheat markets. Ukrainian flour production hardly grew between 2000/2001 and 2001/2002–2002/2003 (APK-Inform, 2004). Flour prices and milling margins gradually declined along with wheat prices (Figure 3), and flour imports did not exceed 2 per cent of total flour supply.

Severe winterkill early in 2003 damaged winter crops and was followed by a prolonged drought in the late spring and early summer, making it likely that Ukraine would become a net importer of food grain in 2003/2004 (von Cramon-Taubadel, 2004: 183). Hence, in the winter and spring of 2003, well before the final size of the wheat harvest was known, prices began to climb. The authorities nevertheless continued to release relatively optimistic harvest forecasts, counteracting price increases and delaying imports. However, once it became clear that less than 5 million tons would be harvested (over 75 per cent less than in the previous two years), prices jumped to new highs in excess of 1,000 UAH/ton.

In response to this 'agricultural crisis', and as in 2000, the government responded with a flurry of *ad hoc* measures. Individual agricultural policy makers at national and regional levels were declared 'responsible' for the shortage of wheat and some were even arrested, investigations were launched into traders' activities and regional authorities were empowered to monitor wheat movements and bread prices. The authorities announced that imports of roughly 2 million tons of wheat at 'reasonable prices' had been negotiated with Russia and Kazakhstan. The prospect of this influx of low-priced wheat (which did not materialise) made private traders wary of importing, thus exacerbating the shortage. Wheat import duties and value-added tax on wheat imports were eventually eliminated, but this took several months. It is rumoured that this delay was partly due to attempts by powerful interests in Parliament to maintain import duties and open tariff rate quotas (in 'suitable' quantities and accessible only to selected traders) instead. Prices finally peaked at over 1,300 UAH/ton after roughly 3 million tons of wheat were imported in the last quarter of 2003 (UkrAgroConsult, 1998–2004). As imports continued and the outlook for the 2004 harvest was promising, wheat prices began to fall, reaching 600–700 UAH/ton by the end of 2004.

The poor wheat harvest in 2003 led to a sharp fall in flour stocks and a consequent rise in flour prices. While the flour price averaged 900 UAH/ton in 2001/2002, it reached 1,800 UAH/ton in June 2003 and peaked close to 2,000 UAH/ton in November 2003 (Figure 3). Flour imports, mainly from Russia and Kazakhstan, increased and in 2003/2004 Ukraine imported 207 thousand tons or 5 per cent of its total flour supply (APK-Inform, 2004). In addition, the State Material Reserve of Ukraine began financing milling and selling flour in large quantities. This 'state' flour, however, was made available only to certain large regional mills and increased uncertainty about flour stocks and prices. Milling margins increased as they were during the 2000/2001 'crisis' and became considerably more volatile (Figure 3). During 2004, flour prices stabilised and gradually decreased.

Overall, Ukrainian wheat and flour markets were characterised by considerable volatility over the period under study. Fluctuations in domestic wheat production twice moved the country from a net export to a net import position and back, shifting domestic wheat and flour prices from export to import parity levels and back as a consequence.[5] These shifts occurred in a policy-making environment in which many participants were either sceptical of or ideologically opposed to free markets, and/or actively involved in rent-seeking. Especially when poor harvests threatened national self-

5 A referee has pointed out that Ukraine exports primarily feed wheat, while we are interested in studying the relationship between flour and milling wheat. Ukraine uses a grading system inherited from Soviet times that defines six wheat classes: Classes I–V are milling wheat, and class VI is feed wheat. Class III is considered the top grade of milling wheat (classes I and II are not produced in Ukraine), followed by class IV. Class V is very rare. Different qualities of wheat are blended to produce lots that fulfil different milling requirements at minimal cost. For example, if the quality of the domestic harvest is low, higher quality lots of domestic feed wheat may be blended with top-quality milling wheat from, for example, Kazakhstan. If the quality of the domestic harvest is high, lower grades of milling wheat may end up in feed. As a result, the distinction between feed and food wheat is not sharp, as there is constant substitution at the margin. This creates a strong link between prices for feed and food classes of wheat in Ukraine, as elsewhere. Hence, although Ukraine primarily exports feed wheat, Figure 2 shows that food wheat (class III) prices in Ukraine have shifted from import to export parity levels and back over time. Figure 3 shows that flour prices have also followed this pattern.

sufficiency in wheat (and hence flour and bread), tentative market-oriented reform would give way to *ad hoc* intervention such as price-fixing and control of grain movements, both across and within Ukraine's borders.

If wheat and flour markets in Ukraine are integrated, there should exist a long-run, co-integrating relationship between their prices. Examination of wheat and flour prices in Figure 3 indicates that they do indeed appear to move together. Hence, the familiar vector error-correction model (VECM) might be an appropriate representation of the underlying data-generating process. However, given the changes in Ukraine's net wheat trade position and the resulting policy volatility discussed above, it is reasonable to hypothesise that this VECM is subject to regime shifts over time. In particular, it is plausible that wheat traders' and millers' behaviour (sales and purchases, contracting, stockholding and ultimately pricing) will have changed in anticipation of and as a result of shifts in Ukraine's net trade position and policy intervention.

## 3. Methods and data

As discussed earlier, a variety of methods for modelling regime-specific behaviour in price transmission analysis have been applied in the literature, all of which involve the use of an observable transition variable. If an appropriate transition variable is available, then clearly it makes sense to incorporate it explicitly when modelling price transmission. For example, if the policy interventions discussed in Section 2 were of the standard 'price wedge' variety, clearly announced and transparently and evenly applied, then it would be relatively straightforward to derive and test hypotheses about how they affected price behaviour. However, the policy measures in question were generally non-tariff in nature, often ambiguously communicated and rarely applied evenly. As outlined in Section 2, different actors (President, Ministry of Agriculture, Parliament) contributed to the agricultural policy mix in Ukraine. There is no convenient record of all the relevant measures (laws, regulations, decrees, threats, etc.), and often no public record at all of whether a measure such as export certification was actually implemented or whether it was applied equally to all traders. For these reasons, constructing a transition variable that effectively measures policy-related shocks would be very difficult, especially after up to seven years have passed.[6] The option of simply using Ukraine's wheat trade as a transition variable must also be rejected. Trade alone would not capture important elements of the policy environment. More fundamentally, weekly trade data are not available to match our weekly price data, which are required to provide sufficient observations for complex modelling approaches.

6 It is conceivable that such an index could be assembled in real time by carefully monitoring the media and government press releases and by regularly interviewing actual market participants.

### 3.1. Markov-switching vector error-correction model

We therefore used the MSVECM to model price transmission between wheat and flour in Ukraine. The key advantage of the MSVECM is that it allows us to model regime-specific behaviour in a setting in which regime shifts are determined by an unobserved state variable that can be modelled as a Markovian process.

The MSVECM is a special case of the general Markov-switching vector autoregressive model. It is essentially a VECM in which some of the parameters shift according to the state of the system. The Markov-switching vector autoregressive model was initially proposed by Hamilton (1989) for analysing the US business cycle. For co-integrated time series, Krolzig *et al.* (2002) and Krolzig and Toro (2001) used the MSVECM to analyse business cycles with a special emphasis on employment. The MSVECM is, however, not restricted to business cycle analysis but provides a general framework for analysing co-integrated times series that follow different regimes whenever the corresponding state variable is not observed. We use the following MSVECM to analyse vertical integration of the markets for wheat and wheat flour in Ukraine:

$$\begin{aligned} \Delta \boldsymbol{p}_t = \boldsymbol{\alpha}_0 + \boldsymbol{\alpha}(s_t)(\boldsymbol{\beta}' \boldsymbol{p}_{t-1}) + D_1(s_t)\Delta \boldsymbol{p}_{t-1} + D_2(s_t)\Delta \boldsymbol{p}_{t-2} \\ + \ldots + D_k(s_t)\Delta \boldsymbol{p}_{t-k} + \boldsymbol{e}_t \end{aligned} \tag{1}$$

where $\boldsymbol{p}_t = (p_t^f, p_t^w)'$ is a vector of market prices for wheat flour (superscript $f$) and wheat (superscript $w$), $\boldsymbol{\alpha}_0$ is a vector of intercept terms, $\boldsymbol{\alpha}$ is a vector of adjustment coefficients, $\boldsymbol{\beta}$ is the co-integrating (long-run equilibrium) vector, $\Delta$ is the first difference operator and $D_1, D_2, \ldots, D_k$ are matrices of short-run coefficients. The vector $\boldsymbol{e}_t$ contains the residual disturbances of the flour and the wheat equations, for which the usual assumptions are made. However, the variance of $e$ is also conditional on the state of the system. The state variable $s_t = 1, \ldots, M$ is an unobserved variable that indicates which of the $M$ possible regimes governs the MSVECM at time $t$. All parameters in the model (except for $\boldsymbol{\alpha}_0$ and $\boldsymbol{\beta}$) are allowed to switch between the regimes. The system might be described as a vector error-correction model with Markov switching in the autoregressive parameters, in the speed of adjustment coefficients and in the variances. Hence, the model allows for regime dependency in the short-run dynamics as well as in the speed of error correction, and for heteroscedasticity.

The most general specification would make the probability of being in state $s_t$ dependent on the entire history of regimes $S_{t-1}$ and on the history of all the variables on the RHS of equation (1), but this general specification would leave the system under-identified. The basic idea of a Markov-switching model is to assume an ergodic Markov process for the probability of observing a certain state, so that the probability of $s_t$ depends only on $s_{t-1}$ and a matrix $\Pi$

of transition probabilities, as follows:

$$\Pr(s_t|S_{t-1}, \Delta p_{t-1}, \Delta p_{t-2}, \ldots, \boldsymbol{\beta}' p_{t-1}, \boldsymbol{\beta}' p_{t-2}, \ldots, ) = \Pr(s_t|s_{t-1}, \Pi). \qquad (2)$$

Each element $\pi_{ij}$ of $\Pi$ gives the transition probability from state $i$ to state $j$. Hence, the sum of each row of $\Pi$ must equal 1, and the number of unknowns in $\Pi$ is thus equal to $M(M-1)$. Since the long-run equilibrium relationship is assumed to be constant over time, the vector $\boldsymbol{\beta}$ does not vary between systems. However, the constant in the co-integrating relationship is allowed to change over time in order to capture regime-dependent changes in the margin.[7]

Estimation of the MSVECM is based on the maximum likelihood principle.[8] The likelihood function is maximised with respect to the parameters in equation (1), a set of parameters corresponding to dummy variables indicating the value of the state variable $s_t$, and the transition probabilities $\Pi_{ij}$. Krolzig (1997) advocated a variant of the Expectation-Maximisation algorithm (Dempster *et al.*, 1977), an iterative procedure that breaks the maximisation down into two steps. First, the state parameters and transition probabilities are estimated conditional on a set of starting values for the coefficients in equation (1). Second, these starting values are updated using the first-order conditions for the maximisation of the likelihood function with respect to the ECM parameters. This sequence is repeated until the state parameters no longer change between subsequent iterations. The estimation procedure is available in the MSVAR package (Krolzig, 2006) for the matrix programming language Ox (Doornik, 2002).

### 3.2. Data

The analysis uses 227 weekly observations from June 2000 to November 2004 (Figure 3) on the average price for class III milling wheat and the wholesale price for top-quality flour in Ukraine, both in natural logarithms.

Since the data are weekly, the sample period considered provides a sufficient number of observations for the chosen econometric approach. At the same time, the sample period is short enough to allow us to assume constant technology in the milling industry. Since the wheat-bread chain is politically sensitive and subject to much *ad hoc* intervention, foreign direct investment in this industry was (and remains) low. Hence, there was no large-scale introduction of new foreign milling technology that might have led to changes in the co-integrating relationship over the sample period. Inflation was moderate in

7 We did not restrict the constant to be part of the co-integrating vector when we estimated the MSVECM. To see how the average margin changes from regime to regime, we took the estimated constant for each VECM regime in the MSVECM and divided it by the corresponding regime-specific estimate of the adjustment coefficient, thus inserting the constant into the co-integrating relationship. Since all prices are in logarithms, these margins are multiplicative.

8 This section only briefly introduces the estimation principles. More details can be found in Hamilton (1989) and Krolzig (1997: Chapter 6).

Ukraine over the relatively short period considered, so we worked with nominal prices.[9]

Over the study period, the average wheat price was 766 UAH/ton. Assuming a flour yield of 75 per cent (1 kg wheat yields 750 g of flour),[10] it took on average 1,021.33 UAH worth of wheat (=766/0.75) to produce 1 kg of flour over this period. Since the average flour price was 1,272 UAH/ton, wheat costs accounted for just over 80 per cent (=1021.33/1272 × 100) of the price of flour. This high cost share and constant technology can be expected to lead to a strong co-integrating relationship between wheat and flour prices. In his seminal article on vertical price transmission, Gardner (1975) derives the following expression for $E_{Px,Pa}$, the elasticity of a processed product price ($P_x$) with respect to the agricultural raw product price ($P_a$):

$$E_{Px,Pa} = \frac{S_a(\sigma + e_b)}{e_b + S_a\sigma - S_b\eta}, \tag{3}$$

where $x$ is the processed product, $a$ is the agricultural raw product, $b$ is the non-agricultural input, $P$ and $S$ are price and cost shares, respectively, $\sigma$ is the elasticity of substitution between $a$ and $b$, $\eta$ is the elasticity of demand for the processed product $x$ and $e_b$ is the supply elasticity of the non-agricultural input $b$.[11] If $S_a \cong 0.8$ for the wheat/flour system, $S_b = (1 - S_a)$ $\cong 0.2$. Since the demand for flour is presumably quite price-inelastic, the last term in the denominator ($S_b\eta$) can be neglected. Furthermore, since the scope for replacing wheat in flour production is limited, $\sigma$ is likely to be very small. Hence, equation (3) simplifies to:

$$E_{Px,Pa} = \frac{S_a e_b}{e_b} = S_a \cong 0.8, \tag{4}$$

which provides an indication of the expected magnitude of the long-run elasticity of flour prices with respect to wheat prices.

9 According to IER (2005), changes in the consumer price index in Ukraine (year on year, end of period) were 2000 (25.8 per cent); 2001 (6.1 per cent); 2002 (−0.6 per cent); 2003 (8.2 per cent); 2004 (12.3 per cent). However, deflating weekly prices would be difficult because the most frequent deflator available is monthly. Hence, any gain from deflating prices would come at the cost of error (e.g. distorted short-run dynamics in the VECM) introduced by whatever arbitrary method was used to derive a weekly from a monthly deflator.

10 A plausible value (see, for example, Canadian Grains Commission, 2007). Data on flour yields in Ukraine are not available.

11 This is equation (19) in Gardner (1975), which refers to adjustment triggered by a change in the raw product price. Equation (18) in Gardner (1975), which refers to adjustment triggered by a change in the processed product price, can also be simplified to approximately $S_a$ under conditions prevailing in the wheat–flour chain.

# 4. Empirical results

## 4.1. Unit root tests

As a prerequisite for the co-integration analysis, we first established the time series properties of the price series using the ADF test, the KPSS test (Kwiatkowski *et al.*, 1992) and a unit root test for processes with level shifts (Lanne *et al.*, 2002), in which the unknown break point is determined by a search over all possible break dates with a sufficiently large lag order. The date that gives the minimal residual sum in the auxiliary regression is chosen. All three tests point to the presence of unit roots in the undifferenced wheat and flour price series (Table 1). However, the tests provide strong evidence in favour of rejecting the null hypothesis of a unit root (or in favour of accepting the null of level stationarity, in the case of the KPSS test) in both the flour and wheat price series in first differences. Hence, the price series appear to display similar I(1) behaviour.

## 4.2. Co-integration analysis and the linear VECM

To test for co-integration, we used the Johansen trace test, based on a reduced rank regression of the vector autoregressive representation with four lags. The null hypothesis of no co-integrating relationship was rejected against the alternative of at least one co-integrating relationship with a $p$-value of less than 0.1 per cent ($\mathrm{LR}^{\mathrm{trace}} = 30.048$), whereas the null that the number of co-integrating vectors is one could not be rejected against the alternative of more than one co-integrating vector ($\mathrm{LR}^{\mathrm{trace}} = 1.922$, $p$-value $= 0.167$). The resulting co-integrating relationship normalised with respect to the flour price is given in equation (5) (standard errors in parentheses):

$$\ln p_t^f = \underset{(0.200)}{1.5976} + \underset{(0.030)}{0.8368} \ln p_t^w + u_t. \tag{5}$$

The estimated value of the elasticity of flour prices with respect to wheat prices (0.837) is close to the expected value of roughly 0.8.

The adjustment coefficients in the corresponding linear VECM are $\alpha^f = -0.127$ (standard error $= 0.026$, $p$-value $<0.1$ per cent) for the flour equation, and $\alpha^w = 0.021$ (standard error $= 0.041$, $p$-value $= 0.603$) for the wheat price equation. Since the deviations from the long-run equilibrium are obtained from the co-integrating vector normalised with respect to the flour price, both $\alpha^f$ and $\alpha^w$ have the expected signs. However, $\alpha^w$ is statistically insignificant. Hence, adjustment towards the long-run equilibrium takes place through changes in flour prices alone, with half of a unit deviation from the long-run equilibrium being corrected within roughly five weeks.

The linear VECM was subjected to diagnostic tests. First, autocorrelation of the disturbances was checked using a vector autocorrelation test up to lag order 12, but was not found.[12] However, homoscedasticity and normality of

12 LM test statistic $= 52.648 < \chi^2_{48,0.05} = 65.17$.

**Table 1.** Results of unit root tests on wheat and flour prices in Ukraine

| Series | Augmented Dickey–Fuller test (Dickey and Fuller, 1979) | | | Unit root test with level shift (Lanne *et al.*, 2002) | | | KPSS: null is stationarity in levels (Kwiatkowski *et al.*, 1992) | | |
|---|---|---|---|---|---|---|---|---|---|
| | Test statistic | Specification | 5% critical value | Test statistic | Specification | 5% critical value | Test statistic | Specification (lag length for Bartlett window) | 5% critical value |
| $\ln p_t^f$ | −1.56 | Six lags, constant | −2.86 | −1.16 | Eight lags, trend, shift dummy | −3.03 | 0.539 | 10 | 0.463 |
| $\ln p_t^w$ | −1.34 | Three lags, constant | | −1.47 | Three lags, trend, shift dummy | | 0.513 | 10 | |
| $\Delta \ln p_t^f$ | −4.45 | Five lags | −1.94 | −4.47 | Seven lags, constant, impulse dummy | −2.88 | 0.203 | 10 | |
| $\Delta \ln p_t^w$ | −6.64 | Two lags | | −6.64 | Two lags, constant, impulse dummy | | 0.151 | 10 | |

*Source:* Own calculations with data in Figure 3.
*Note:* Lag length for the KPSS statistic is approximated according to $l_8 = \mathrm{int}(8(t/100)^{1/4})$.

the disturbances were rejected. The full White test for vector heteroscedasticity yielded a test statistic of 268.35 ($>\chi^2_{105,0.05} = 129.9$), and the non-normality test rejected the null hypothesis of normality with a $p$-value of less than 0.01. Fat tails in the distribution of the residuals and heteroscedasticity could both be caused by parameter instability. Hence, the system was checked for stability using a Chow breakpoint test of the null hypothesis that all parameters in the system remain constant over time against the alternative that all coefficients except $\beta$ and the residual covariance matrix change. The Chow test statistic is asymptotically distributed as $\chi^2$; however, since the actual distribution under the null is non-standard (Candelon and Lütkepohl, 2001), a bootstrap procedure implemented in JmulTi (http://www.jmulti.org) was used to calculate empirical $p$-values for different breakpoints. Figure 4 shows the results of some parameter stability tests for the linear VECM. Panel 'a' contains the bootstrapped $p$-values of the Chow breakpoint test (every second week was used as a possible breakpoint). The $p$-values for most of the breakpoints lie well below 0.05. Hence, the linear VEC system appears to be affected by structural breaks, and the representation of the price movements on Ukrainian wheat flour and wheat markets using a time-invariant ECM is inappropriate.

The Chow breakpoint test maintains the hypothesis of a stable long-run relationship. However, as pointed out by an anonymous referee, the long-run relationship between the flour and the wheat prices might also be subject to structural change. Although we considered this unlikely (because without fundamental changes in the market structure, this relationship is largely determined by technological parameters that will not have changed much over the five-year sample period), we tested the stability of the co-integrating vector using the recursively calculated $\tau$-test statistic of Hansen and Johansen (1999). This test evaluates the stability of the recursively estimated eigenvalues. Strong changes in the eigenvalues would suggest instability in the co-integrating vector, which might originate from the speed of adjustment coefficient $\alpha$, the slope coefficient $\beta$, or both. We calculated the test statistic based on the concentrated likelihood, which seems more suitable than recursive estimation of the short-run parameters because we are interested specifically in the stability of the co-integrating vector. The results are presented in Figure 4, panel 'b'. Since the $\tau$-test statistic is less than the critical value (1.36) over the entire sample period, there appears to be no parameter instability in the co-integrating relationship.

Recursive instability tests with unknown breakpoints have limited power when there are structural changes in the data (Juselius, 2006). Hence, we also used a likelihood ratio test proposed by Hansen (2003) for the stability of the slope coefficient in the co-integrating relationship based on multiple, known breakpoints. We set these at mid-2001 and mid-2003, when changes in Ukraine's net trade position for wheat occurred (Figure 3). No deterministic terms were included so that the null hypothesis of no structural change is tested against the alternative of regime-specific differences in the slope coefficient in equation (5). The null of no structural changes between the regimes could not

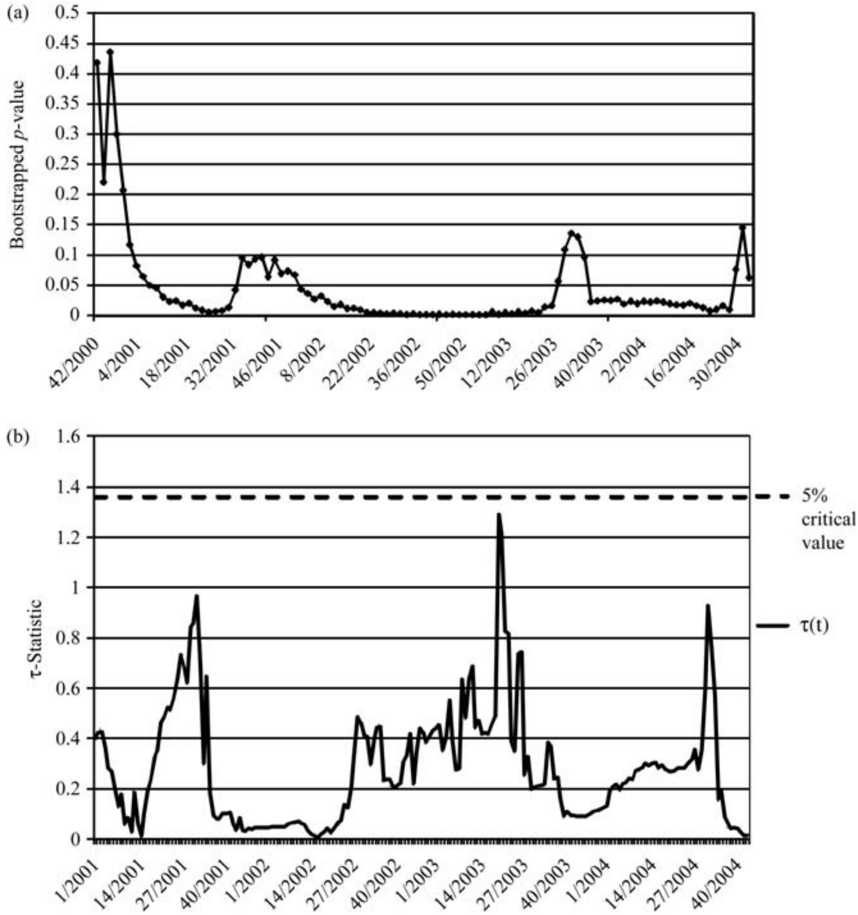


**Figure 4.** Parameter stability tests of the linear VECM. (a) Bootstrapped Chow breakpoint test $p$-values for the wheat/flour price VECM (based on 500 replications). (b) Recursive $\tau$-statistic, starting from 2001, week 1.
*Source:* Own calculations with data in Figure 3.

be rejected.[13] Since both tests with known and unknown breakpoints failed to produce evidence of parameter instability, we used the estimated constant co-integrating relation in equation (5) for further analysis in an MSVECM.

### 4.3. Markov-switching vector error-correction model

The number of lags and the number of regimes in the MSVECM were selected according to the Akaike information criterion (AIC). A formal test for $M$ regimes against the alternative of $M + 1$ regimes is difficult because several

13 Test statistic $= 6.85 < \chi^2_{4,0.05} = 9.49$.

parameters in the unrestricted model are not identified under the null hypothesis, leading to a non-standard distribution of the usual likelihood-based test statistics. Nevertheless, the AIC strongly favours a specification with four regimes and three lags [MS(4) − VECM(3)].[14] The residual diagnostics for this specification indicate that autocorrelation, heteroscedasticity and non-normality are not present.

The estimated parameters of this MSVECM specification are presented in Table 2. An interesting feature is the lower adjustment coefficients compared with the simple VECM. The significant coefficient in the flour equation is reduced by factors of 3 (regimes 1–3) and 6 (regime 4). The adjustment coefficient in the wheat equation remains statistically insignificant in all regimes. Another intriguing feature is the distinct variation in the residual standard errors ($\sigma_\varepsilon^f$ and $\sigma_\varepsilon^w$) between the regimes. Further characterisation of the regimes in terms of price levels ($p^f$ and $p^w$) and multiplicative margins ($p^f/p^w$) is summarised together with the deviation from the long-run equilibrium in Table 3.

The prices for both wheat and flour are highest in regime 1, followed by regime 4. Regimes 2 and 3 show even lower average prices. The observed average margins increase continuously from regime 1 to regime 4. Comparing these observed margins with those implied by the estimated long-run relationship, we see that the difference between the margin implied by the estimation outcome and the observed margin is lowest for regime 1. This is also evident in the corresponding average value of the error-correction term in this regime, which is very close to zero, indicating that there is on average little deviation from the long-run equilibrium. Regime 2 displays somewhat larger differences between observed and estimated margins, but it is only in regimes 3 and 4 that these differences become visible. Speed of adjustment, residual standard errors and the resulting intercept in the long-run relation allow for a more detailed interpretation of the single regimes[15]:

Regime 1 ('normal integration') is characterised by relatively small values for the residual standard errors $\sigma_\varepsilon^f$ and $\sigma_\varepsilon^w$. Both the logarithm of the intercept (=1.560) and the speed of adjustment parameter in the flour equation ($\alpha^f = -0.041$) are at their normal levels. The estimated margin is 1.59, which is identical to the expected flour/wheat price ratio based on the long-run cost share (0.8368), and an input requirement of 1.33 tons of wheat per ton of flour (i.e. 0.75 kg of flour from 1 kg of wheat).

14 Psaradakis and Spagnolo (2003) recommended the AIC for determining the number of regimes. Smith *et al.* (2006) found in their Monte-Carlo simulations that the AIC has a tendency to retain too many regimes. In the context of a MS autoregressive model, they proposed a Markov-Switching Criterion (MSC) that approximates the Kullback–Leibler distance. It is an open question whether this criterion is also superior in models containing exogenous regressors, such as an error correction term. In our case, the MSC criterion also favours the specification with four regimes.

15 Information on regime-specific impulse-response functions is available from the authors on request. Impulse-response behaviour in a Markov-switching model is complex, because regime-specific behaviour is modified by the probability of switching to other regimes, each with its own regime-specific behaviour.

**Table 2.** The estimated MS(4)-VECM(3) for wheat and flour prices in Ukraine

| Variable | Regime 1 | | Regime 2 | | Regime 3 | | Regime 4 | |
|---|---|---|---|---|---|---|---|---|
| | $\Delta P^f$ | $\Delta p^w$ | $\Delta p^f$ | $\Delta p^w$ | $\Delta p^f$ | $\Delta p^w$ | $\Delta p^f$ | $\Delta p^w$ |
| Constant | 0.063* (0.011) | −0.013 (0.021) | 0.063* (0.011) | −0.013 (0.021) | 0.063* (0.011) | −0.013 (0.021) | 0.063* (0.011) | −0.013 (0.021) |
| $\Delta p^f_{t-1}$ | −0.030 (0.034) | −0.299* (0.055) | 0.236* (0.064) | −0.173 (0.206) | 0.233* (0.092) | 0.110* (0.051) | 0.240 (0.221) | 0.877* (0.417) |
| $\Delta p^f_{t-2}$ | 0.051* (0.026) | 0.043 (0.043) | 0.097 (0.057) | −0.035 (0.177) | −0.124 (0.117) | 0.051 (0.061) | 0.789* (0.251) | 0.592 (0.452) |
| $\Delta p^f_{t-3}$ | −0.221* (0.030) | −0.295* (0.048) | 0.001 (0.035) | −0.224* (0.111) | 0.163 (0.094) | 0.075 (0.057) | 0.431 (0.366) | 1.054 (0.662) |
| $\Delta p^w_{t-1}$ | 0.293* (0.063) | 0.619* (0.105) | −0.039* (0.018) | 0.038 (0.056) | 0.308* (0.107) | 0.329* (0.061) | −0.351 (0.220) | −0.999* (0.421) |
| $\Delta p^w_{t-2}$ | −0.017 (0.011) | −0.208* (0.018) | 0.048 (0.035) | 0.410* (0.121) | −0.361* (0.109) | 0.096 (0.064) | 1.275* (0.361) | 1.450* (0.661) |
| $\Delta p^w_{t-3}$ | 0.044 (0.032) | 0.432* (0.054) | 0.068 (0.040) | 0.508* (0.130) | 0.222* (0.056) | −0.026 (0.032) | 0.160 (0.395) | 2.941* (0.725) |
| $\mathrm{ECT}_{t-1}$ | −0.041* (0.007) | 0.008 (0.013) | −0.041* (0.007) | 0.004 (0.013) | −0.044* (0.007) | 0.014 (0.013) | −0.026* (0.010) | −0.027 (0.018) |
| $\sigma_\varepsilon^{f/w}$ | 0.0034 | 0.0056 | 0.0052 | 0.0170 | 0.0229 | 0.0123 | 0.0368 | 0.0631 |
| Constant restricted in the ECT term | 1.560 | | 1.530 | | 1.449 | | 2.418 | |

*Source*: Own calculations with data in Figure 3 using MSVAR for Ox (Doornik, 2002; Krolzig, 2006).
*Note*: Standard errors in parentheses.
*Statistical significance at 1%.

**Table 3.** Characterisation of the estimated regimes

| Indicator | Regime | | | |
|---|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| Average flour price | 1352.5 | 1241.9 | 1194.3 | 1323.5 |
| Average wheat price | 837.4 | 743.6 | 704.3 | 773.5 |
| Average observed margin | 1.62 | 1.67 | 1.70 | 1.71 |
| Average estimated margin | 1.59 | 1.57 | 1.46 | 3.79 |
| Average error-correction term | 0.0288 | 0.0670 | 0.1582 | −0.7862 |

*Source:* Own calculations based on data in Figure 3 and estimation results.

Regime 2 ('calming') exhibits increased residual variance (relative to regime 1, $\sigma_\varepsilon^f$ and $\sigma_\varepsilon^w$ increase 1.5- and 3-fold, respectively), while the intercept and $\alpha^f$ remain at normal levels. The estimated margin is virtually unchanged; however, since the observed margin in this period has increased, there is downward pressure on the margin.

Regime 3 ('alert') is characterised by a strong increase in the variability of the errors for flour ($\sigma_\varepsilon^f$ and $\sigma_\varepsilon^w$ increase by factors of 7 and 2, respectively, relative to regime 1). The logarithmic intercept is slightly reduced (=1.449), and $\alpha^f$ (= −0.044) remains almost unchanged. The observed margin is now substantially higher than the estimated margin, so that the average error-correction term is positive. Regime 3 has the lowest prices; however, the actual margin is still relatively high, thus indicating that there is upward pressure on wheat prices.

Regime 4 ('disarray') has the highest residual standard errors ($\sigma_\varepsilon^f$ and $\sigma_\varepsilon^w$ increase 10- and 12-fold, respectively, relative to regime 1). The logarithm of the intercept is very high (=2.418), and the speed of adjustment in the flour price change equation is almost halved compared with regime 1 ($\alpha^f = -0.026$). The estimated margin is very high, far above the maximum margin observed in the data (2.5). In consequence, deviations from a long-run equilibrium (the error-correction term) in this regime are negative. This exerts downward pressure on wheat prices and upward pressure on flour prices in order to bring the observed and the estimated margins in line with one another.

However, the dynamics of the system depend not only on the regime parameters but also on the transition dynamics. Table 4 presents the matrix of transition probabilities from regime $s_{t-1}$ to regime $s_t$, the average frequency and the average duration of each regime. The values on the main diagonal indicate the probability of persistence in a regime. Regimes 1 and 2 are found to be the most persistent. Together, these two regimes account for about two-thirds of all observations. While regimes 1 and 2 both last for about 4 weeks on average, regimes 3 and 4 last only 2 and 1.5 weeks on average, respectively. From both regime 1 and 2, if a change takes place then regime 3 is the most likely outcome in the subsequent period

**Table 4.** Transition matrix and regime properties for the MS(4)-VECM(3)

| From regime | To regime | | | |
|---|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| 1 | 0.736 | 0.067 | 0.142 | 0.055 |
| 2 | 0.121 | 0.728 | 0.151 | 0.000 |
| 3 | 0.117 | 0.236 | 0.528 | 0.119 |
| 4 | 0.194 | 0.069 | 0.370 | 0.367 |
| Duration | 3.79 | 3.68 | 2.12 | 1.58 |
| Number of observations | 73.7 | 74.5 | 59.3 | 17.6 |

*Source:* Own calculations based on estimation results.
*Note:* The number of observations in each regime is calculated by multiplying the ergodic regime probabilities with the sample size.

(probabilities of 14 and 15 per cent, respectively). From regime 3, the system either calms down (to regime 2, 24 per cent; to regime 1, 12 per cent), or uncertainty in the market culminates in disarray (regime 4, 12 per cent). The estimated regime probabilities indicate that 59 observations are attributed to regime 3. From regime 4, calming either takes place via regime 3 as an intermediate step (37 per cent), or directly to regime 1 (19 per cent). However, regime 4 is not very persistent: only 18 observations are attributed to this regime, and it displays only a 37 per cent probability of persistence.

Figure 5 presents more information on the duration of each regime. The graph plots on the vertical axis the cumulative probability that a regime

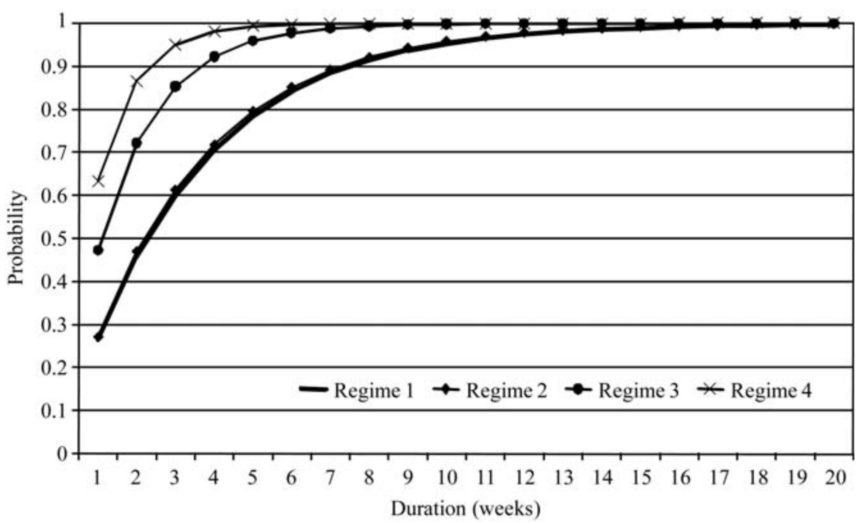


**Figure 5.** Regime duration: the cumulative probability of exiting regime $s_t$ after $t$ weeks.
*Source:* Own calculations based on results in Table 2.

will not last longer than *t* weeks (measured on the horizontal axis), i.e. that the regime will end after at most *t* weeks. Regimes 1 and 2 follow virtually identical routes, while regime 3 is substantially shorter lived. Regime 4 (disarray) is the least persistent: the probability that regime 4 will have ended after 3 weeks is greater than 95 per cent.

## 5. Discussion

We next relate the results of the MSVECM estimation to the market and policy developments discussed in Section 2. In Figure 6, the evolution of Ukrainian wheat and flour prices is contrasted with the classification of the individual observations into the four regimes. Figure 6 also shows the maximum smoothed probability for the most probable regime in each period. We see that the maximum smoothed probabilities are for the most part close to 1, indicating that most observations are attributed clearly to a specific regime. Of the four regimes identified above, regime 4 is the most interesting, because the lack of adjustment and the inflated margin and uncertainties in the price equations imply substantial welfare losses and re-distribution in the Ukrainian wheat–flour chain. Regimes 1 and 2 in particular are more prevalent and persistent than regime 4, but they correspond to phases in which wheat and flour prices are relatively stable and error correction is relatively strong, in other words wheat–flour price transmission appears to be functioning well. By comparison, the parameters in regime 4 point to a considerable reduction in price transmission, and it is interesting

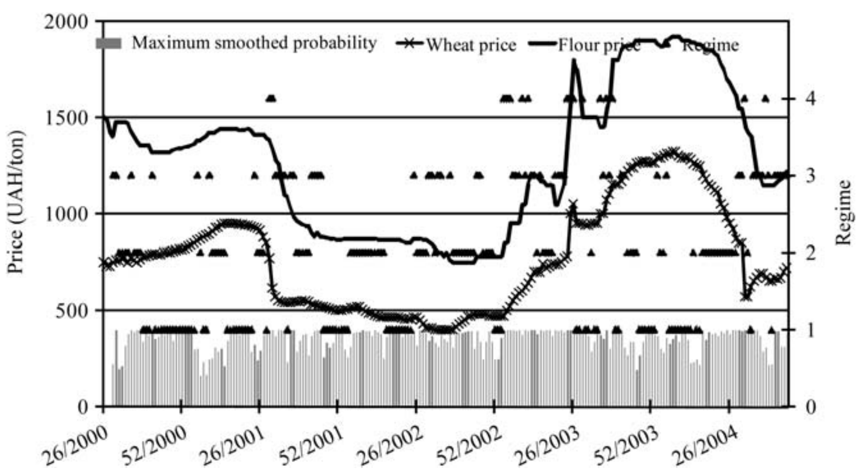


**Figure 6.** Flour and wheat prices in Ukraine (UAH/ton, left axis), classification of the observations into regimes (right axis) and smoothed probability for the most likely regime (right axis), July 2000–November 2004.
*Source:* Own calculations with data in Figure 3 and results in Table 2.

to consider what might cause this reduction. Hence, in the following we focus on regime 4.

Figure 6 indicates that there were distinct episodes of regime 4. The peak probabilities of observing regime 4 coincide with periods of major shocks to flour and wheat markets in Ukraine. The first peak occurs in the second half of July 2001, when a bumper crop returned Ukraine from a net import to a net export situation for wheat, and the price for wheat dropped more rapidly than the price for wheat flour, leading to an above-average margin (Figure 3). The next peak occurs in January 2003, when the first news of severe winterkill in the Ukrainian wheat crop emerged. The subsequent one-period occurrences of regime 4 in the 9th and 11th weeks of 2003 can be attributed to the same cause and may be related to conflicting information and rumours about the full extent of the damage from winterkill and the likelihood of policy responses.

Later in 2003, additional episodes of regime 4 can be observed, which coincide with policy interventions on the markets for wheat and flour. The first episode occurs in the summer and begins with a block of three weeks duration (weeks 24–26 of 2003), followed by a one-period observation in week 29. These dates coincide with heavy political activity: on June 29 and July 24, important cabinet resolutions were issued, which set out the intended government reaction to the low harvest and the emerging shortage of wheat. Regime 4 is observed two weeks in advance of these dates, which might be due to rumours and intense public discussion preceding the official resolutions. The interventionist character of many of the proposed measures, such as allowing for regional administrative control of grain shipments, or regulating bread prices, destabilised markets, as signalled by the high probability of regime 4.

Following a brief ‘calming’ of four weeks duration, regime 4 re-occurred in the fall of 2003. Official announcements of low-priced wheat imports from Kazakhstan and Russia coincide with episodes of regime 4 in weeks 35, 37 and 39. Beginning in October 2003, grain markets and policies in Ukraine began to stabilise as imports began to enter the country.

Figure 6 shows that much of 2003 can be viewed as a period in which wheat and flour markets were destabilised (regime 3 or 4). It also indicates that after some stabilisation in the first half of 2004, the prevalence of regimes 3 and 4 increased again in July 2004. These episodes of instability might be attributable to the presidential election campaign in Ukraine, in which both leading candidates explicitly referred to the regulation of wheat trade and bread price controls in their election programmes.

Causality between market instability and policy intervention can run in both directions, of course, and our analysis does not permit us to determine empirically which directions applied when. The *ad hoc* and frequently uncoordinated nature of the policy interventions outlined in Section 2 and anecdotal evidence provided by market participants like traders over the years strongly suggest, however, that policy intervention frequently exacerbated market imbalances.

## 6. Conclusions

This paper analyses vertical market integration between wheat and flour in Ukraine over the years 2000–2004. Interference on these markets by policy makers, especially in response to changes in Ukraine's wheat trade balance, was varied, sometimes intense and often contradictory, leading to considerable uncertainty and making it difficult for private actors like farmers, traders and millers to plan consistently.

We therefore hypothesised that transmission between wheat and flour prices in Ukraine was not constant over the period. This suspicion was heightened by the results of a Chow breakpoint test that rejected the null hypothesis of parameter constancy in a linear VECM involving wheat and flour prices. Since no observable variable signalling regime changes was available for the case of wheat and flour markets in Ukraine, we used the MSVECM, a method that is new to the price transmission literature. A MSVECM with three lags and four regimes, estimated using 228 weekly observations of wheat and flour prices in Ukraine between June 2000 and November 2004, was found to perform well.

The empirically identified four regimes can be interpreted as different conditions governing the price relationship between flour and wheat at different points in time. Differences in the residual standard errors, the milling margin and the speed of adjustment to disequilibrium distinguish the regimes from one another. The most imprecise regime, i.e. the one with the highest residual variance and most volatile margins, is regime 4. The recurrence of this regime over time can be linked to market shocks such as changes in Ukraine's wheat trade balance, and to policy reactions triggered by these shocks. In particular, in 2003 and the latter part of 2004, *ad hoc* policy interventions coincide with high probabilities of regime 4. Although causality can run both ways, this suggests that much policy intervention in response to shocks to Ukraine's wheat and flour markets may have increased rather than reduced instability.

The MSVECM approach employed here is a useful addition to the set of methods that can be used to study regime-specific behaviour in price transmission. Specifically, it can generate interesting insights in settings where a transition variable that determines regime shifts (such as the size of the error-correction term in threshold error-correction models) is not readily available. Applications to other settings and further investigation of the robustness of the results of MSVECM estimation (number of regimes, number of lags) would help to increase the confidence with which results such as those presented here can be interpreted in the context of actual market developments. Of course, where a transition variable is available and indicated by theoretical considerations – for example in spatial settings where transaction costs are hypothesised to generate threshold error-correction behaviour – the corresponding alternative model can be expected to outperform the MSVECM. However, MSVECM modelling can nonetheless provide a useful complement to alternative approaches whenever the theoretical underpinning for a transition variable is weak and/or this variable cannot be measured reliably.

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## References

APK-Inform (2004). Ukrainian grain market. Weekly Analytical and Statistical Report. Kyiv, Ukraine: APK Inform, July.

Bakucs, L. Z. and Fertő, I. (2005). Marketing margins and price transmission on the Hungarian pork market. In S. Brosig and H. Hockmann (eds), *How Effective Is the Invisible Hand? Agricultural and Food Markets in Central and Eastern Europe*. Studies on the Agricultural and Food Sector in Central and Eastern Europe, Vol. 31, Halle: Institute of Agricultural Development in Central and Eastern Europe (IAMO): 134–149.

Barrett, C. (2001). Measuring integration and efficiency in international agricultural markets. *Review of Agricultural Economics* 23: 19–32.

Barrett, C. and Li, J. (2002). Distinguishing between equilibrium and integration in spatial price analysis. *American Journal of Agricultural Economics* 84: 292–307.

Baulch, B. (1997). Transfer costs, spatial arbitrage and testing for food market integration. *American Journal of Agricultural Economics* 79: 477–487.

Berkowitz, D. and DeJong, D. N. (2000). Economic consequences of Russia's internal border. In P. Wehrheim, E. V. Serova, K. Frohberg and J. V. Braun (eds), *Russia's Agro-food Sector: Towards Truly Functioning Markets*. Boston: Kluwer Academic Publishers, 185–201.

Bojnec, Š. and Peter, G. (2002). Price transmission and marketing margins in the Slovenian beef and pork markets during transition. In *Paper Presented at the 10th EAAE Congress 'Exploring Diversity in the European Agri-food System'*, 28–31 August 2002, Zaragoza.

Canadian Grain Commission (2007). Wheat, No. 1 Canada western red spring quality data for 2007 harvest survey grade composite samples compared to 2006 and 1997–06 mean. http://www.grainscanada.gc.ca/Quality/Wheat/2007/CWRS_1_2007_e.pdf. Accessed 21 August 2008.

Candelon, B. and Lütkepohl, H. (2001). On the reliability of Chow type tests for parameter constancy in multivariate dynamic models. *Economics letters* 73: 155–160.

Chavas, J.-P. and Mehta, A. (2004). Price dynamics in a vertical sector: the case of butter. *American Journal of Agricultural Economics* 86: 1078–1093.

von Cramon-Taubadel, S. (ed.) (2001). Price determination and government policy on Ukrainian grain markets. In S. Cramon-Taubadel, S. von Zorya and L. Striewe (eds), *Policies and Agricultural Development in Ukraine*. Aachen: Shaker Verlag, 20–31.

von Cramon-Taubadel, S. (2004). The 2003 wheat harvest: Crisis! What crisis? In S. Cramon-Taubadel, S. von Demyanenko and A. Kuhn (eds), *Ukrainian Agriculture – Crisis and Recovery*. Aachen: Shaker Verlag, 183–200.

von Cramon-Taubadel, S., Elsner, E., von der Malsburg, E, Movchan, V. and Nivyevskiy, O. (2008). Ukraine. In K. Anderson and J. Swinnen (eds), *Distortions to Agricultural Incentives in Europe's Transition Economies*. Washington, DC: World Bank, 175–218.

Dempster, A. P., Laird, N. M. and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm (with discussion). *Journal of the Royal Statistical Society B* 39: 1–38.

Demyanenko, S. and Zorya, S. (2004). Taxation and Ukrainian agriculture. In: S. von Cramon-Taubadel, S. Demyanenko and A. Kuhn (eds), *Ukrainian Agriculture: Crisis and Recovery*. Aachen: Shaker Verlag, 25–39.

Dickey, D. A. and Fuller, W. A. (1979). Estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association* 74: 427–431.

Doornik, J. A. (2002). *Object-Oriented Matrix Programming Using Ox*. 3rd edn. London: Timberlake Consultants Press and Oxford: www.nuff.ox.ac.uk/Users/Doornik. Accessed 25 April 2009.

Fackler, P. L. and Goodwin, B. K. (2001). Spatial price analysis. In B. Gardner and G. Rausser (eds), *Handbook of Agricultural Economics*, Vol. 1B. Amsterdam: Elsevier, 971–1024.

Gardner, B. L. (1975). The farm-retail price spread in a competitive food industry. *American Journal of Agricultural Economics* 57: 399–409.

Gonzalo, J. and Pitarakis, J.-Y. (2006). Threshold effects in cointegrating relationships. *Oxford Bulletin of Economics and Statistics* 68: 813–833.

Goodwin, B. K., Grennes, T. J. and McCurdy, C. (1999). Spatial price dynamics and integration in Russian food markets. *Policy Reform* 3: 157–193.

Goodwin, B. K., Harper, D. and Schnepf, R. (2000). Short-run demand relationships in the US fats and oils complex. In *Paper Presented at the NCR-134 Conference on Applied Commodity Analysis, Forecasting and Market Risk Management*, 17–18 April 2000, Chicago, IL.

Goodwin, B. K. and Piggott, N. E. (2001). Spatial market integration in the presence of threshold effects. *American Journal of Agricultural Economics* 83: 302–317.

Gregory, A. W. and Hansen, B. E. (1996). Residual-based tests for cointegration in models with regime shifts. *Journal of Econometrics* 70: 99–126.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57: 357–384.

Hansen, B. E. (1992). Tests for parameter instability in regressions with I(1) processes. *Journal of Business and Economic Statistics* 10: 321–335.

Hansen, P. R. (2003). Structural changes in the cointegrated vector autoregressive model. *Journal of Econometrics* 114: 261–295.

Hansen, H. and Johansen, S. (1999). Some tests for parameter constancy in cointegrated VAR-models. *The Econometrics Journal* 2: 306–333.

IER (Institute for Economic Reform and Policy Consulting) (2005). *Monthly Economic Monitor Ukraine*, No. 12-2005. Kiev. http://ierpc.org/ierpc/memu/memu_62_dec_05_ua.pdf. Accessed 25 April 2009.

Juselius, K. (2006). *The Cointegrated VAR Model: Methodology and Applications*. New York: Oxford University Press.

Krolzig, H.-M. (1997). *Markov-Switching Vector Autoregressions – Modelling, Statistical Inference, and Application to Business Cycle Analysis*. Berlin: Springer-Verlag.

Krolzig, H.-M. (2006). MSVAR – an Ox package designed for the econometric modelling of univariate and multiple time series subject to shifts in regime. Version 1.32a: http://www.krolzig.co.uk/msvar.html. Accessed 25 April 2009.

Krolzig, H.-M. and Toro, J. (2001). *A New Approach to the Analysis of Business Cycle Transitions in a Model of Output and Employment.* Department of Economics discussion paper series No. 59, Oxford: University of Oxford.

Krolzig, H.-M., Marcellino, M. and Mizon, G. E. (2002). A Markov-switching vector equilibrium correction model of the UK labour market. *Empirical Economics* 27: 233–254.

Kuhn, A. (2000). Regional (dis-)integration of Russia's agro-food markets. In P. Wehrheim, E. V. Serova, K. Frohberg and J. von Braun (eds), *Russia's Agro-food Sector: Towards Truly Functioning Markets*. Boston: Kluwer Academic Publishers, 203–219.

Kwiatkowski, D., Phillips, P. C. B., Schmidt, P. and Shin, Y. (1992). Testing the null of stationarity against the alternative of a unit root: how sure are we that the economic time series have a unit root? *Journal of Econometrics* 54: 159–178.

Lanne, M., Lütkepohl, H. and Saikkonen, P. (2002). Comparison of unit root tests for time series with level shifts. *Journal of Time Series Analysis* 23: 667–685.

Loy, J.-P. and Wehrheim, P. (1999). Spatial food market integration in Russia. In G. H. Peters and J. von Braun (eds), Food Security, Diversification, and Resource Management: Refocusing the Role of Agriculture. *Proceedings of the 23rd International Association of Agricultural Economists (IAAE) Conference in Sacramento*. Aldershot, Hants, UK: Ashgate Publishing, 421–431.

McNew, K. and Fackler, P. L. (1997). Testing market equilibrium: is cointegration informative? *Journal of Agricultural and Resource Economics* 22: 191–207.

Meyer, J. (2004). Measuring market integration in the presence of transaction costs – a threshold vector error correction approach. *Agricultural Economics* 31: 327–334.

Meyer, J. and von Cramon-Taubadel, S. (2004). Asymmetric price transmission: a survey. *Journal of Agricultural Economics* 55: 581–611.

Psaradakis, Z. and Spagnolo, N. (2003). On the determination of the number of regimes in Markov-switching autoregressive models. *Journal of Time Series Analysis* 24: 237–252.

Quandt, R. E. (1958). The estimation of the parameters of a linear regression system obeying two separate regimes. *Journal of the American Statistical Association* 53: 873–880.

Rozelle, S. and Swinnen, J. F. M. (2004). Success and failure of reform: insights from the transition of agriculture. *Journal of Economic Literature* 42: 404–456.

Sephton, P. S. (2003). Spatial market arbitrage and threshold cointegration. *American Journal of Agricultural Economics* 85: 1041–1046.

Serra, T., Gil, J. M. and Goodwin, B. K. (2006). Local polynomial fitting and spatial price relationships: price transmission in EU pork markets. *European Review of Agricultural Economics* 33: 415–436.

Smith, A., Naik, P. A. and Tsai, C. L. (2006). Markov-switching model selection using Kullback–Leibler divergence. *Journal of Econometrics* 134: 553–577.

Striewe, L. and von Cramon-Taubadel, S. (1999). Die Getreideproduktion der Ukraine: Verpasste Chancen und dringender Handlungsbedarf. In S. von Cramon-Taubadel and L. Striewe (eds), *Die Transformation der Landwirtschaft in der Ukraine: Ein weites Feld*, Kiel: Wissenschaftsverlag Vauk Kiel KG, 74–87.

Teräsvirta, T. (2004). Smooth transition regression modeling. In H. Lütkepohl and M. Krätzig (eds), *Applied Econometric Time Series*. New York: Cambridge University Press, 222–242.

UkrAgroConsult (weekly). *Agrinews Weekly*. Kyiv.

USDA(a) (monthly). *Grain: World Markets and Trade*. Foreign Agricultural Service Circular Series, United States Department of Agriculture. http://www.fas.usda.gov/archive.asp. Accessed 20 August 2008.

USDA(b) (monthly). *Wheat Outlook*. Economic Research Service, United States Department of Agriculture. http://www.ers.usda.gov/publications/outlook. Accessed 20 August 2008.

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